st

arr

10	4	2	7	9	12	3
0	1	2	3	4	5	6

$O(n^2)$

$O(n)$

ans

12	7	7	9	12	-1	-1
0	1	2	3	4	5	6

6
5

index

arr[st.top()]

```
for(i=0 ; i<n ; i++)
{
    while(!st.empty() && arr[i] > arr[st.top()])
    {
        ans[st.top()] = arr[i];
        st.pop();
    }

    st.push(i)

}
```

NSR

0	1	2	3	4	5	6
5	8	6	7	3	2	9

0	1	2	3	4	5	6
3	6	3	3	2	-1	-1

9

2

12 10 8

12	10	7	8	2	4	5
0	1	2	3	4	5	6

8
10
12

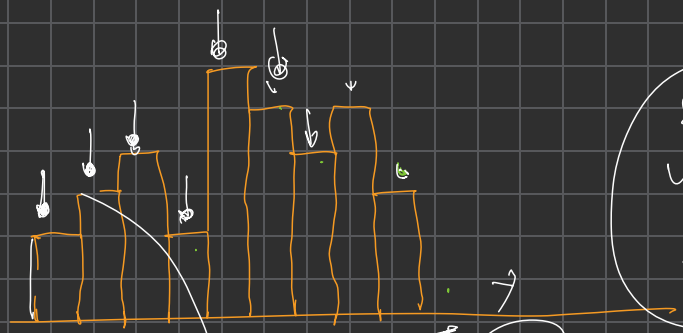
-1	12	10	10	8	8	8
0	1	2	3	4	5	6

0	1	2	3	4	5
2	10	4	3	8	5

10	-4	8	8	10	10

6

6
2
2



n =

2	3	4	2	6	5	4	5	9
0	1	2	3	4	5	6	7	8

9	3	3	9	5	6	8	8	9

-1	0	1	-1	3	3	3	6	3

$2 \times 9 = 18$

$4 \times 4 = 16$

$3 \times 5 = 15$

$\parallel$

18 ✓

$3 - 0 - 1$

$2 \times 3 = 6$

$9 - (-1) - 1$

$\Rightarrow 9 * 2 = 18$ ✓

18 ✓

NSR - NSL - 1 * height

$\parallel$

area ✓